MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCGEO100 - THE EARTH: AN INTRODUCTION

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome (CO) to enhance their overall outcome attainability.

Part A

Attend any 7 of the following in 60 words

Each question carries **2** marks

1. Define Geomorphology. Mention any two geomorphological features [U][CO1]

2. Differentiate outer core and inner core. [U][CO1]

3. How do tectonic movements influence endogenic processes? [U][CO2]

4. List out the main factors that control metamorphism. [U][CO3]

5. How does the physical and biological weathering processes accelerate the chemical weathering processes? [U][CO2]

6. Define carbonization with respect to process of fossilisation. [K][CO4]

7. Distinguish between apparent dip and true dip. [U][CO5]

8. How does weathering contribute to the rock cycle? [U][CO3]

(7x2 marks =14 marks)

Part B

Write short notes on any 3 of the following in 250 words

Each question carries **8** marks

9. Discuss the composition and characteristics of earth's crust including its types. [U][CO1]

10. Discuss the rock cycle, and how does it explain the transformation of igneous, sedimentary, and metamorphic rocks?

[U][CO3]

11. Compare and contrast fold and fault in rocks with neat diagrams. [U][CO5]

12. Differentiate Palynology & Ichnology, Micropalaeontology & Palaeontology with examples

[U][CO1]

13. Compare the rate of soil formation in tropics, temperate, arid and polar regions. [U][CO4]

(3x8 marks = 24 marks)

Part C

Write essay on any 1 of the following in 400 words.

Each question carries **12** marks

14. Compare and contrast the lithosphere, asthenosphere, and lower mantle in terms of their structure, composition, and their roles within the Earth system

[U][CO1]

15. Describe igneous, metamorphic and sedimentary rocks and their process of formation.

[U][CO4]

(1x12 marks =12 marks)